

Step-1: Create storage account .

Step-2: Create a catalog- retail_catalog_02
Schema -> sales_analytics

The screenshot shows the Databricks Catalog Explorer interface. On the left, a sidebar lists various Databricks components like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, and SQL Warehouses. The main panel displays the 'retail_catalog_02' catalog. Under the 'Catalog' section, a list of schemas is shown: default, information_schema, sales_analytics, retail_catalog_vaishnavi, sdatta, vaishnavi-catalog, data_engineer, retail_proj2_sdatta_cat, sdatta-catalog, tiasa_catalog1, tiasa_catalog2, tiasa_catalog4, tiasacatalog3, vp07, and vp07_capstone. The 'retail_catalog_02' catalog is selected, and its details are shown on the right. The 'Overview' tab is active, showing a description, a search bar for schemas, and a table of schemas. The table has columns for Name, Owner, and Created at. The schemas listed are default, information_schema, and sales_analytics. The 'About this catalog' section shows the owner as Tredence LabsKraft23. The 'Tags' section has an 'Add tags' button.

Name	Owner	Created at
default	tredence23@michaelcloudkraft...	Aug 26, 2025, 10:22 AM
information_schema	System user	Aug 26, 2025, 10:22 AM
sales_analytics	tredence23@michaelcloudkraft...	Aug 26, 2025, 10:30 AM

Step-3: Grant required permission to the catalog.

The screenshot shows the Databricks Catalog Explorer interface, specifically the 'Permissions' tab for the 'retail_catalog_02' catalog. The left sidebar is the same as in the previous screenshot. The main panel shows the 'retail_catalog_02' catalog selected. The 'Permissions' tab is active, showing a table of permissions. The table has columns for Principal, Privilege, and Object. The permissions listed are: Tredence LabsKraft23 (ALL PRIVILEGES), vaishnavi.maurya@tredence.com (EXTERNAL USE SCHEMA), nebip46416@evoxury.com (BROWSE), nebip46416@evoxury.com (USE CATALOG), nebip46416@evoxury.com (USE SCHEMA), vmaurya24@gmail.com (BROWSE), vmaurya24@gmail.com (EXECUTE), vmaurya24@gmail.com (READ VOLUME), vmaurya24@gmail.com (SELECT), vmaurya24@gmail.com (USE CATALOG), vmaurya24@gmail.com (USE SCHEMA), and All account users (BROWSE). The 'Grant' button is visible at the top left of the permissions table.

Principal	Privilege	Object
Tredence LabsKraft23	ALL PRIVILEGES	retail_catalog_02
vaishnavi.maurya@tredence.com	EXTERNAL USE SCHEMA	retail_catalog_02
nebip46416@evoxury.com	BROWSE	retail_catalog_02
nebip46416@evoxury.com	USE CATALOG	retail_catalog_02
nebip46416@evoxury.com	USE SCHEMA	retail_catalog_02
vmaurya24@gmail.com	BROWSE	retail_catalog_02
vmaurya24@gmail.com	EXECUTE	retail_catalog_02
vmaurya24@gmail.com	READ VOLUME	retail_catalog_02
vmaurya24@gmail.com	SELECT	retail_catalog_02
vmaurya24@gmail.com	USE CATALOG	retail_catalog_02
vmaurya24@gmail.com	USE SCHEMA	retail_catalog_02
All account users	BROWSE	retail_catalog_02

Step-4: Ingest data into bronze->silver->gold.

```
# retail_sales_pipeline.py
import dlt
from pyspark.sql.functions import *

from pyspark.sql.types import *
sales_schema = StructType([
    StructField("sale_id", IntegerType(), True),
    StructField("product_id", IntegerType(), True),
    StructField("region", StringType(), True),
    StructField("store_id", StringType(), True),
    StructField("quantity", IntegerType(), True),
    StructField("sales_amount", DoubleType(), True),
    StructField("sale_date", TimestampType(), True)
])
products_schema = StructType([
    StructField("product_id", StringType(), True),
    StructField("product_name", StringType(), True),
    StructField("category", StringType(), True),
    StructField("price", DoubleType(), True),
    StructField("discount_rate", DoubleType(), True)
])
# ----- CONFIG -----
RAW_SALES_PATH      = "abfss://raw@vaishnavidata.dfs.core.windows.net/Sales/"
RAW_PRODUCTS_PATH   = "abfss://raw@vaishnavidata.dfs.core.windows.net/Products/"

# Schema/history locations (must be writable by the pipeline cluster)
SCHEMA_LOCATION_SALES      = "/dbfs/dlt_schema/retail/sales"      # DBFS path
example
SCHEMA_LOCATION_PRODUCTS   = "/dbfs/dlt_schema/retail/products" # DBFS path
example
```

```

# Target Unity Catalog (catalog.schema) for tables
TARGET_SCHEMA = "retail_catalog_02.sales_analytics"

# Helper to produce full UC table names
def tbl(name):
    return f"{TARGET_SCHEMA}.{name}"

# ----- BRONZE (raw ingestion) -----
@dlt.table(
    name=tbl("bronze_sales"),
    comment="Bronze: raw sales ingested with Auto Loader (no transforms)",
    table_properties={"quality": "bronze"}
)
def bronze_sales():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .option("header", "true")
            .schema(sales_schema)
            # .option("inferSchema", "true") # avoid
relying on inference here
            .option("cloudFiles.schemaLocation", SCHEMA_LOCATION_SALES)
            # .option("cloudFiles.schemaEvolutionMode", "addNewColumns") #
allow new columns to appear
            .load(RAW_SALES_PATH)
    )

@dlt.table(
    name=tbl("bronze_products"),
    comment="Bronze: raw product metadata ingested with Auto Loader (no
transforms)",
    table_properties={"quality": "bronze"}
)
def bronze_products():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "json") # change to 'csv' if your
product files are CSV

```

```

        .option("header", "true")
        .schema(products_schema)
        .option("cloudFiles.schemaLocation", SCHEMA_LOCATION_PRODUCTS)
        # .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
        .load(RAW_PRODUCTS_PATH)
    )
@dlt.table(
    name="silver_sales",
    comment="Cleaned sales data (Silver)"
)
@dlt.expect_or_drop("non_null_orderid", "sale_id IS NOT NULL")
@dlt.expect("valid_quantity", "quantity >= 0")
def silver_sales():
    return (
        dlt.read("bronze_sales")
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .filter(col("sale_id").isNotNull())
    )

@dlt.table(
    name="silver_products",
    comment="Cleaned products data (Silver)"
)
@dlt.expect_or_drop("non_null_productid", "product_id IS NOT NULL")
def silver_products():
    return (
        dlt.read("bronze_products")
        .filter(col("product_id").isNotNull())
    )

# =====
# ♦ GOLD (Business Aggregates)
# =====
@dlt.table(
    name="gold_daily_revenue",
    comment="Daily revenue per region (Gold)"
)
def gold_daily_revenue():

```

```

return (
    dlt.read("silver_sales")
    .groupBy("region", "sale_date")
    .agg(
        sum(col("sales_amount")).alias("daily_revenue") #  clean alias
    )
)

@dlt.table(
    name="gold_product_performance",
    comment="Aggregated product performance (Gold)"
)
def gold_product_performance():
    return (
        dlt.read("silver_sales")
        .join(dlt.read("silver_products"), on="product_id", how="left")
        .groupBy("product_id", "product_name", "category")
        .agg(
            sum("quantity").alias("total_units_sold"),
            count("sale_id").alias("total_orders")
        )
    )

```

Microsoft Azure databricks

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Jobs & Pipelines >

26_aug_vaishnavi ☆ Send feedback

Development Production Settings Schedule Share Start

Run: 8/26/2025, 2:13:12 PM · Completed Select tables for refresh

Graph List

Streaming table: bronze_products Completed · 12s

Materialized view: silver_products Completed · 9s

Materialized view: gold_product_perf... Completed · 10s

Streaming table: bronze_sales Completed · 9s

Materialized view: silver_sales Completed · 10s

Materialized view: gold_daily_revenue Completed · 11s

Details Expectations Schema Flows

Type: Streaming table

Source code: /Users/tredence23@michaelkoudkraft@hotmail.onmicrosoft.com/Untitled Notebook 2025-08-26 11:44:03

Table name: retail_catalog_02.sales_analytics.bronze_products

Status: Completed

Start time: 8/26/2025, 2:25:12 PM

Duration: 12s

Comment: Bronze: raw product metadata ingested with Auto Loader (no transforms)

View logs

After adding remaining files to container the records increase.



Added this file as record no. 5001 with extra columns without adding new columns.

A	B	C	D	E	F	G	H	I	J
sale_id	product_id	region	store_id	quantity	sales_amount	sale_date	Promotio_code	hello_code	fivetrancode
1	739	US	Store_12	14	150.45	8/12/2025 15:52	kesh40		90 uy6_poskjbksbv

	1 ² sale_id	1 ² product_id	A ^B region	A ^B store_id	1 ² quantity	1.2 sales_amount	📅 sale_date
4992	992	532	EU	Store_43	16	1036.55	2025-08-12T22:38:
4993	993	31	EU	Store_37	10	1634.36	2025-08-08T15:20:
4994	994	60	MEA	Store_4	18	309.23	2025-08-03T22:31:
4995	995	100	EU	Store_5	12	1700.15	2025-08-12T08:43:
4996	996	113	US	Store_25	9	210.93	2025-08-04T19:32:
4997	997	11	LATAM	Store_22	12	2160.63	2025-08-23T21:19:
4998	998	170	LATAM	Store_25	7	631.87	2025-08-20T03:37:
4999	999	921	EU	Store_20	12	153.28	2025-08-22T20:09:
5000	1000	719	APAC	Store_44	19	3138.18	2025-08-25T20:05:
5001	1	739	US	Store_12	14	150.45	null
5002	null	null	null	null	null	null	null
5003	null	null	null	null	null	null	null

Pipeline event log details

✕

This event is associated with `retail_catalog_02.sales_analytics.bronze_sales`.

Summary

JSON

✖ ERROR

2025-08-26 15:57:59 IST

flow_progress

Failed to resolve flow: 'retail_catalog_02.sales_analytics.bronze_sales'.

🔗 Diagnose error

Error details

📄

```

py4j.protocol.Py4JJavaError: Traceback (most recent call last):
py4j.protocol.Py4JJavaError: An error occurred while calling o953.load.
: com.databricks.sql.cloudfiles.errors.CloudFilesIllegalArgumentException: Schema evolution mode addNewColumns is
not supported when the schema is specified. To use this mode, you can provide the schema through `cloudFiles.sche
maHints` instead.
    at com.databricks.sql.cloudfiles.errors.CloudFilesErrors$.addNewNotSupported(CloudFilesErrors.scala:115)
    at com.databricks.sql.cloudfiles.options.CloudFilesOptionsBase.assertSchemaEvolutionModeSupported(CloudFi
lesOptionsBase.scala:335)
    at com.databricks.sql.cloudfiles.options.CloudFilesOptionsBase.<init>(CloudFilesOptionsBase.scala:195)
    at com.databricks.sql.fileNotification.autoIngest.CloudFilesSourceOptions.<init>(CloudFilesSourceOptions.
scala:54)
    at com.databricks.sql.fileNotification.autoIngest.CloudFilesSourceProvider.sourceSchema(CloudFilesSourceP
rovider.scala:84)
    at org.apache.spark.sql.execution.datasources.DataSource.sourceSchema(DataSource.scala:274)
    at org.apache.spark.sql.execution.datasources.DataSource.sourceInfo$lzycompute(DataSource.scala:158)

```

I hardcoded the schema so on schema evolution as add new column so error is coming.

2 minutes ago (11s) 1 SQL

```
%sql
select * from retail_catalog_02.sales_analytics.bronze_sales
```

▼ (2) Spark Jobs
 ▶ Job 14 [View](#) (Stages: 1/1)
 ▶ Job 15 [View](#) (Stages: 1/1)

▶ `_sqldf: pyspark.sql.connect.dataframe.DataFrame = [sale_id: integer, product_id: string ... 10 more fields]`

		sale_date	_rescued_data	Promotio_code	hello_code	fivetrancode	sale_amt
1	0.45	8/12/2025 15:52	null	kesh40	90	uy6_poskjbksbv	null
2	null	null	null	null	null	null	null
3	null	null	null	null	null	null	null
4	null	null	null	null	null	null	null
5	null	null	null	null	null	null	null
6	null	null	null	null	null	null	null

Extra columns added after adding:

```
.option("cloudFiles.schemaEvolutionMode", "addNewColumns")
    .option("cloudFiles.schemaHints", "sale_id INT, region STRING,
sales_amount DOUBLE")
```

To the def bronze_sales()

Versioning

